St. Philomena's College (Autonomous), Mysore

PG Department of Studies and Research in Chemistry

Question Bank (Revised Curriculum 2018 onwards)

First Year- Second Semester (2019 -21 Batch)

Course Title (Paper Title):

Quantum, Nuclear, PhotoChemistry and Molecular Spectroscopy QP Code:55104

Unit	SI. No	Questions	Marks
1	1.	Write the de-Broglie relation and explain the terms involved in it.	2
1	2.	State Heisenberg's uncertainty principle.	2
1	3.	Give an example for commutative and non commutative operators.	2
1	4.	Write a note on Hamiltonian operator.	2
1	5.	Verify the operator δ^2 is linear.	2
1	6.	What are the conditions for a function to be called as Class -Q function.	2
1	7.	Define Eigen values and Eigen functions with an example.	2
1	8.	Write a Schrodinger wave equation for a particle trapped in 1-D box.	2
1	9.	State Variation theorem.	2
2	10.	Calculate the wavelength of an alpha particle having mass 6.6×10^{-27}kg moving with a speed of 10^5cm/sec.	2
1	11.	Show that Sin2X is not an Eigen function of the operator d/dx	2
1	12.	Prove that Eigen values of a Hermitian operator are real.	2
2	13.	Define half life and write a relation between half life and decay constant.	2
	14.	What is Chain disintegration give example.	2
2	15.	Write an examples for alpha, beta, gamma decays.	2
2	16.	Write a decay scheme for an isotope $^{226}Ra_{88}$	2
2	17.	Define average life period .	2
2	18.	Write decay kinetics equation and explain the terms involved in it.	2
2	19.	Define average life period .	2
2	20.	Define atomic mass unit.	2

2	21.	Define isotopes with example	2
2	22.	Define Mirror nuclei and isotone with examples	2
2	23.	What is Magic numbers.	2
2	24.	Define mass loss of a radioactive isotope.	2
2	25.	Define radioactive units Becquerel and curie	2
2	26.	Write the difference between transient and secular equilibrium,	2
2	27.	How to determine the half life for a mixture of independent half lives	2
2	28.	Define packing fraction.	2
2	29.	Define Binding energy and binding energy per nucleon.	2
2	30.	State Beer- Lamberts law and write its mathematical form	2
2	31.	Write the significance of term symbol.	2
2	32.	Define stability belt based on N/Z ratio.	2
2	33.	Define dead time and coincidence loss.	2
3	34.	Write the principle of G.M Counter.	2
3	35.	Write the selection rule for rotation of diatomic molecules	2
3	36.	Write the condition for the moment of inertia for linear molecules.	2
3	37.	Write the condition for the moment of inertia for symmetric top molecules.	2
3	38.	Write the condition for the moment of inertia for prolate molecules.	2
3	39.	Write the condition for the moment of inertia for oblate molecules.	2
3	40.	Write the selection rule for rotation of non rigid rotator.	2
3	41.	Write the expression for moment of inertia for polyatomic molecules.	2
3	42.	Define stark effect in rotation spectra.	2
3	43.	Write the equation for the energy of a anharmonic oscillator	2
3	44.	Write the different modes of vibrations of carbon dioxide.	2
3	45.	write the expression for fundamental frequency in terms of force constant.	2

3	46.	Write the condition for the moment of inertia for asymmetric top molecules.	2
3	47.	Define larmour frequency	2
3	48.	Write a note on Zeeman splitting.	2
3	49.	Identify the given sample esr active or not Be and Be^{+}	2
3	50.	Identify the given sample esr active or not N_2 and O_2	2
3	51.	Write the value of g for unpaired electron and define spin -spin relaxation process	2
3	52.	Write the difference between spin-spin and spin lattice relaxation process.	2
3	53.	Define Kramer's degeneracy and zero field splitting.	2
3	54.	Write the hyperfine splitting pattern for CH_2 radical.	2
3	55.	Write the hyperfine splitting pattern for 1,4- semi benzoquinone.	2
2	56.	Define quantum yield and mention its significance.	2
2	57.	Write the factors affecting quantum yield	2
2	58.	State Grothus-drapper law	2
2	59.	State Einstein law of photochemical equivalence.	2
2	60.	Define chain length and chain inhibition.	2
2	61.	Write the quantum yield and wavelength range for acetone and diethyl ketone actinometers.	2
2	62.	Define photosensitization. Give example	2
2	63.	Define quenching and quencher	2
2	64.	Define the phenomena of fluorescence	2
2	65.	Write the applications of fluorescence	2
2	66.	Write an expression for photochemical rate law for the decomposition of acetaldehyde.	2
2	67.	Write the expression for photochemical rate law for the formation of hydrogen and chlorine.	2
2	68.	What is the effect of photo degradation on COD value.	2

2	69.	Write the principle involved in photo degradation.	2
2	70.	Define primary and secondary process in photochemical reaction.	2
2	71.	Write the different steps involved in photochemical reaction with example.	2
2	72.	Define dead time of a GM Counter	2
1	73.	What is Heisenberg Uncertainty principle? Mention its consequences.	2
1	74.	Enumerate the need for approximate methods in quantum chemistry.	2
2	75.	Define Photonuclear reaction with an example.	2
3	76.	Mention the factors that govern the intensities of rotational spectra in a rigid diatomic rotator.	2
3	77.	State stark effect in microwave spectra	2
2	78.	Explain the principles of actinometers.	2
2	79.	The half life of Ra-226 is 1600yrs.Calculate decay constant and the value of one curie.	2
2	80.	Calculate the energy of one Einstein of a light of wavelength 3600°	2
2	81.	Give two examples for specific and photonuclear reactions.	2
1	82.	Calculate debroglie wavelength for an electron and predict its nature	2
3	83.	What is hyper fine coupling	2
1	84.	Differentiate between linear and laplacian operators	2
2	85.	Mention the factors affecting quantum yield	2
3	86.	Find the number of vibration mode of the non linear molecule with 7 atoms.	2
3	87.	Calculate the rotational constant and energy of HCN molecule at the first excited state whose frequency is 28.453/cm and I= 1.023×10^{-46}Kg	2
3	88.	Differentiate between Harmonic and Anharmonic oscillator	2
3	89.	When does a molecule on an atom show ESR spectrum.	2
2	90.	S.T the binding energy and nuclear stability are related.	2
3	91.	State why a molecule need to have a permanent dipole moment to be microwave active?	2

2	92.	Write the Bethe's notation for the general nuclear reaction.	2
2	93.	Define Photosensitization.	2
3	94.	Why carbon dioxide has two absorption peaks in its IR Absorption spectrum.	2
2	95.	What are term symbols	2
3	96.	Define Kramer’s degeneracy and zero field splitting.	2
1	97.	outline class-Q functions	2
3	98.	Calculate the ESR signal for benzene and anthracene molecules	2
1	99.	Write the wave equation for hydrogen atom and explain the terms	2
2	100.	Define binding energy and packing fraction.	2
2	101.	Give the reasons for high quantum yield for the photochemical formation of HCl	2
2	102.	Define the terms dead time and coincidence loss	2
2	103.	Define half life and mean life of radioactive isotopes	2
2	104.	Write a note on decay kinetic equation and explain the terms involved in it.	2
3	105.	Give the gross and specific selection rules for microwave spectra.	2
3	106.	What is force constant? Give its significance.	2
3	107.	Write the possible vibrational modes of CO_2 molecules and mention their IR activity	2
2	108.	How does half life of a radio active nucleus is determined.	2
2	109.	What is term symbol? Give an example.	2
2	110.	What is meant by COD	2
3	111.	Give an example for each symmetric and asymmetric top molecules.	2
3	112.	How many number of vibrational degree of freedom present in CH_4 molecule.	2
2	113.	Write the difference between acetone and diethyl ketone actinometers	2
3	114.	What are symmetric top molecules.	2

1	115.	Write the difference between operator and operand.	2
2	116.	What do you mean by spallation reaction? Give one example.	2
2	117.	Give the significance of GM plateau	2
1	118.	Differentiate between Eigen values and Eigen functions.	2
2	119.	How do you account for the activation in quantum yield? Give examples.	2
2	120.	State Rutherford's theory of rate of decay	2
2	121.	Give the principle of Szilard-Chalmers process	2
3	122.	Write the possible vibrational modes of CO_2 molecules and mention their IR activity	2
2	123.	calculate the weight of 2milli curie of ^{14}C($t_{1/2}$ of ^{14}C= 5720yrs)	2
3	124.	What is Fermi resonance	2
3	125.	How many number of vibrational degree of freedom present in CH_4 molecule.	2
2	126.	Write the difference between acetone and diethyl ketone actinometers	2
3	127.	What are symmetric top molecules.	2
1	128.	Define Laplacian operator and give its significance	2
2	129.	Define radio active decay	2
3	130.	How fundamental bands are different from overtone bands	2
3	131.	What is meant by hyperfine coupling	2
2	132.	A 0.1mg of sample of pure $^{231}Pu_{91}$ undergoes 1.04×10^7 disintegration per minute. Calculate its half life	2
3	133.	Give reason why H_2 and CH_4 do not give rotation spectra while HCl and CH_3Cl do.	2
1	134.	Define degeneracy and write wave equation for hydrogen atom	2

2	135.	Write the term symbols for ^{19}F and ^{12}C.	2
3	136.	How many fundamental bands are observed in the infrared spectrum of benzene.	2
3	137.	State the total number of vibrational modes in triatomic linear and symmetric molecules.	2
1	138.	Write the difference between classical and quantum mechanics.	2
3	139.	State which of the following are microwave active OCS,CO_2,N_2 and HF.	2
	140.		
1	141.	Derive a Schrodinger wave equation for a particle trapped in 1-D box.	5
1	142.	Derive a Schrodinger wave equation for a particle trapped in 3-D box.	5
1	143.	State and variation theorem.	5
1	144.	Discuss the application of Schrodinger wave equation for Hydrogen atom.	5
1	145.	By using the Schrodinger wave equation deduce an equation for stretched string.	5
1	146.	Explain the application of Schrodinger wave equation for rigid rotator.	5
1	147.	Discuss the application of Schrodinger wave equation for simple harmonic oscillator..	5
1	148.	Discuss the application of perturbation theory and apply this to obtain wave function of a perturbed system with first order correction only	5
1	149.	Discuss the quantum mechanics of Harmonic oscillator.	5
1	150.	Write a note on Eigen values, Eigen functions and class-Q functions.	5
1	151.	Write a note on Laplacian and Hamiltonian operator	5
1	152.	Set up the Schrodinger wave equation for hydrogen atom and solve Θ	5
1	153.	Write the postulates of Quantum mechanics.	5
2	154.	Write a note on Szilard-Chalmers process and Spallation reactions.	5
2	155.	Give the brief survey of alpha, beta, gamma decays.	5
2	156.	Write a note on nuclear stability based on N/Z ratio	5
2	157.	Write a note on Mass loss, Binding energy and binding energy per nucleon.	5

2	158.	Calculate the binding energy/nucleon of $2He_4$	5
2	159.	Derive a general equation for the growth of a radioactive daughter. A-----B-----C with decay constants λ_1 and λ_2	5
2	160.	Explain briefly about G.M Counters.	5
2	161.	What is photonuclear and specific nuclear reactions.	5
2	162.	Obtain an expression for parent-daughter decay growth relationship.	5
2	163.	Expain different types of nuclear reactions based on the nature of bombarding particles.	5
2	164.	Derive a photochemical rate law for the photochemical combination of hydrogen and chlorine	5
2	165.	Derive a photochemical rate law for the photochemical decomposition of acetaldehyde	5
2	166.	Write a note on potassium ferrioxalate actinometers.	5
2	167.	Explain about unaryl oxalate actinometer and the extent of decomposition of oxalic acid titrated against $KMnO_4$	5
2	168.	Derive a term symbols for atomic carbon and fluorine.	5
2	169.	Write a note on G. M. Plateau and dead time.	5
2	170.	Discuss the mechanism involved in photo catalytic process	5
2	171.	Explain the applications of ZnO in photo degradation of dyes and pesticides.	5
3	172.	Explain the phenomenon in which you can determine the precise atomic mass by microwave spectroscopic technique.	5
3	173.	The first line of the rotation spectrum of CO is observed at 3.8424/cm, calculate the bond length.	5
3	174.	Explain the applications of ESR spectroscopy	5
3	175.	How do you obtain a simple ESR spectra for an inorganic molecule.	5
3	176.	How do you calculate precise atomic mass of ^{13}C isotope in $^{12}CO_{16}$ molecule from microwave spectroscopy	5
2	177.	Write a note on photo catalysts ZnO and TiO_2.	5

3	178.	Deduce an expression for moment of inertia for a linear triatomic molecules.	5
3	179.	Explain the instrumentation technique involved in microwave spectrometer.	5
3	180.	Explain the rotation spectra of diatomic molecules.	5
3	181.	Explain the rotation spectra of a rigid rotator.	5
3	182.	Explain the effect of isotopic substitution on rotation spectra.	5
3	183.	Write a brief note on relative intensities of spectral lines.	5
3	184.	Explain the vibrational-rotational spectra of carbon monoxide	5
3	185.	Obtain an expression for fundamental frequency and overtones for a simple harmonic oscillator.	5
3	186.	Sketch the hyperfine splitting pattern for benzene and naphthalene radicals.	5
3	187.	Calculate the esr frequency of an unpaired electron in a magnetic field of 3000guass	5
3	188.	Calculate the g value if the methyl radical shows esr at 3290G in a spectrometer operating at 9230MHz.	5
3	189.	Calculate the esr frequency of unpaired electron in a magnetic field of 0.33T given g=2 and bohr magneton is 9.273×10^{-24}	5
3	190.	In esr spectrometer, the esr frequency of free electron is 9.5GHz. Calculate the magnetic field at which spectrometer is operated.	5
2	191.	What is an actinometer? Describe the functioning and application of Uranyl oxalate actinometers.	5
2	192.	Discuss the decomposition of acetaldehyde.	5
3	193.	Sketch the symmetry of the water molecule and its three fundamental vibrations and also separate them into parallel and perpendicular mode of vibrations.	5
3	194.	With suitable example explain the mode of separation polyatomic molecules by making use of the principle of moment of inertia	5
2	195.	One gram of radioactive element decays by beta emission to 0.125g in 200hrs. How much more time will elapse until only 0.10g of it is left.	5
2	196.	Write a note on Effect of photo degradation on COD value and g value and factors affecting it.	5

2	197.	Obtain an expression for the determination of dead time of a GM counter.	5
3	198.	Explain the factors affecting group frequency in vibration spectroscopy.	5
2	199.	Obtain a relation between parent and daughter nuclei when half life of parent is greater than daughter.	5
3	200.	Explain rotation spectra of OCS molecule.	5
2	201.	What is binding energy? Calculate the binding energy/ nucleon in helium atom which has a mass of 4.00260 a.m.u, mass of one neutron= 1.008665 a.m.u and mass of hydrogen atom = 1.007825a.m.u	5
2	202.	How is nuclear stability can be explained on the basis of binding energy and packing fraction.	5
2	203.	Write a note on dissociation of H_2 molecule by Photosensitization.	5
3	204.	How do you determine the bond length and moment of inertia from rotational spectra?	5
3	205.	Explain the rotation spectra of polyatomic molecules-CH_3F	5
2	206.	The half life of ^{22}Na is 2.6yr. 89% of it decays by β^+ emission and the rest by electron capture. Calculate Partial decay constant and half life for each mode.	5
2	207.	Why ZnO/TiO_2 acts as photo catalyst in photo degradation of organic compounds?	5
2	208.	The moment of inertia of CS2 molecule is $2.56\times 10^{-38} g.cm^2$. Calculate the C-S bond length.	5
3	209.	Explain the effect of hydrogen bonding and ring strain on vibration spectra.	5
2	210.	Define fluorescence. Explain the characteristics and mechanism of quenching of fluorescence.	5
3	211.	What are symmetric top and asymmetric top molecules? Explain their activity towards microwave irradiations?	5
2	212.	Obtain the rate expression for the photochemical formation of HCl	5
1	213.	State and prove variation theorem.	5
2	214.	How do you determine g value? and what are the factors affecting the magnitude of g value.	5

2	215.	Explain the general characteristics of radioactive decay	5
2	216.	Differentiate between specific nuclear and photonuclear reactions.	5
2	217.	Define quantum yield? Explain briefly the factors affecting quantum yield.	5
2	218.	What is meant by dead time ? Explain how it is determined.	5
2	219.	Discuss the photo degradation of a dye in the presence of TiO_2.	5
3	220.	Discuss the experimental technique of microwave spectrometer.	5
2	221.	Explain parallel and perpendicular vibrations with suitable examples.	5
3	222.	Outline the application of ESR spectroscopy for simple organic free radicals.	5
3	223.	What is g factor? List and explain the factors affecting the g value.	5
3	224.	The first line in the pure rotational spectrum of $^{1}H\ ^{35}Cl$ appears at 21.18/cm. Calculate bond length of the molecule, given atomic masses of H and Cl are 1.008a.m.u and 35.45 a.m.u respectively.	5
1	225.	Set up the Schrodinger equation for H-atom	5
1	226.	What are operators? Explain commutative and noncom mutative operators with examples.	5
1	227.	Discuss the applications of Schrodinger equation to harmonic oscillator.	5
1	228.	Write a note on Eigen values and eigen functions and Bohr's theory of Hydrogen atom.	5
2	229.	Discuss the kinetics of photochemical decomposition of CH_3CHO	5
2	230.	Discuss the production of radio-isotopes and labeled compounds by bombardment process.	5
2	231.	What is meant by dead time of the GM counter? Explain the determination of dead time.	5
2	232.	Explain neutron activation analysis	5
2	233.	Discuss the oppenheimer - phillips process	5
2	234.	Discuss Photonuclear reactions.	5
2	235.	Explain the resonance fluorescence and quenching of fluorescence	5

Q.P Code: 14MSCBH140

St. Philomena's College (Autonomous) Mysore

M.Sc. (Chemistry) II Semester Final Examination May 2016

Subject: PHYSICAL CHEMISTRY

Title: Quantum, Nuclear & Photochemistry, Microwave and Infrared Spectroscopy

Time: 3 Hours **Max Marks: 70**

Instruction to the Candidates: Question paper has 2 parts answer both the parts. Provide clear and concise answer to the questions. Use chemical equations / drawings to support your answer where ever appropriate.

PART –A

Answer the following questions: **10x2=20**

1. a. Define Eigen value and Eigen function.
   b. Calculate the de Broglie wave length for an electron and predict the nature of it (Mass of electron = 9.1095×10^{-31}kg).
   c. What is Laplacian and Hamiltonian Operator? Give example for each.
   d. Obtain Radioactive decay law equation.
   e. Write Bethe's rotation with example.
   f. Write some of the factors affecting Quantum yield.
   g. What are Term Symbols?
   h. Calculate the notational constant and energy of HCN molecule at the first excited state whose frequency is $28.453cm^{-1}$ and moment of inertia is 1.023×10^{-46}kg.
   i. Illustrate symmetric and asymmetric top molecules.
   j. Define Kramers degeneracy.

PART –B

Answer any FIVE of the following: **5x10=50**

2. a. Obtain an expression for the Heisenberg Uncertainty principle and write its significances. 4+6
   b. How do you arrive at a wave equation for a stretched strings.
3. a. Equate Schrodinger equation to a harmonic Oscillator. 5+5
   b. State and prove variation theorem.

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4. a. Write a note on nuclear stability or radioactive nuclides. 4+4+2

   b. Obtain relationship between parent and daughter when half-life of parent is greater than daughter.

   c What is photonuclear reaction? Give an example.

5. a. Explain photochemical kinetics of decomposition of CH_3CHO. 5+5

   b. Write a note on Uranyloxalate and potassium ferrioxalate actinometrs.

6. a. List out the selection rule in microwave spectroscopy. 4+4+2

   b. Explain rotation spectra of OCS molecule.

   Define dead time of a G.M.Counter

7. a. How do you obtain a simple ESR spectra for an inorganic molecule. 5+5

   b. Obtain vibration – rotation spectra of carbon monoxide.

8. a. How you calculate precise atomic mass of ^{13}C isotope in $^{12}CO^{16}$ molecule from microwave spectroscopy. 4+3+3

   b. Where you come across Oppenheimer Phillip process? Explain.

   c. What is first order and second order correction?

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